Assignment 5 Report

Vision Transformer Fine-tuning & Adversarial Robustness

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WandB Q-1: run/assignment5 | **WandB Q-2:** run/assignment5_q2
Docker: Yes

1. Q1: ViT-S Fine-tuning on CIFAR-100

1.1 Objective

Question 1 studies transfer learning with a ViT-S model pre-trained on ImageNet applied to CIFAR-100. Required experiments include (i) head-only fine-tuning, (ii) LoRA on attention qkv layers, (iii) a full LoRA grid over rank and alpha, and (iv) hyperparameter selection with Optuna.

1.2 Experimental Summary

The baseline fine-tunes only the classification head. LoRA experiments keep the head trainable while injecting low-rank adapters into qkv projections. The required grid sweeps ranks $\{2, 4, 8\}$ and alphas $\{2, 4, 8\}$ at dropout 0.1. A separate Optuna search optimises dropout beyond the fixed grid.

1.3 Q1 Baseline

Table 1: Q1 head-only baseline summary.

Metric	Value
Best Val Acc.	0.8190
Test Acc.	0.8143
Test Loss	0.6215
Trainable Params	38 500

Table 2: Q1 baseline epoch metrics.

Ep.	Tr. Loss	Va. Loss	Tr. Acc	Va. Acc
1	2.2098	0.9253	0.5139	0.7598
2	1.2480	0.7470	0.6791	0.7898
3	1.1063	0.6895	0.7054	0.8046
4	1.0559	0.6595	0.7145	0.8072
5	1.0087	0.6468	0.7252	0.8110
6	0.9893	0.6356	0.7297	0.8132
7	0.9648	0.6285	0.7349	0.8176
8	0.9323	0.6235	0.7457	0.8150
9	0.9232	0.6149	0.7464	0.8190
10	0.9110	0.6119	0.7498	0.8178

1.4 Q1 LoRA Grid

Table 3: Required LoRA grid (dropout = 0.1).

r	α	Val Acc.	Test Acc.	Train.%
2	2	0.9012	0.8999	0.346
2	4	0.9020	0.9017	0.346
2	8	0.9020	0.9020	0.346
4	2	0.9010	0.8994	0.514
4	4	0.9042	0.8997	0.514
4	8	0.9058	0.9012	0.514
8	2	0.9016	0.9016	0.850
8	4	0.9042	0.9017	0.850
8	8	0.9054	0.9012	0.850

1.5 Best Q1 LoRA (Optuna)

The best Optuna configuration (r=8, α=4, dropout $\approx$ 0.2962) achieved a validation accuracy of **0.9070** and test accuracy of **0.9049** — a **+9.06 pp** gain over the baseline while keeping fewer than **1%** of parameters trainable.

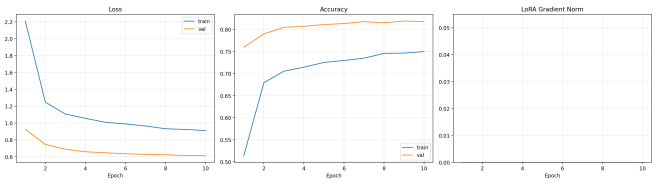
Table 4: Best Optuna LoRA epoch metrics.

Ep.	Tr. Loss	Va. Loss	Tr. Acc	Va. Acc
1	1.5463	0.4334	0.6480	0.8754
2	0.5856	0.3610	0.8306	0.8886
3	0.5106	0.3358	0.8500	0.8918
4	0.4645	0.3219	0.8618	0.8976
5	0.4346	0.3173	0.8701	0.9012
6	0.4082	0.3044	0.8782	0.9042
7	0.3908	0.3073	0.8820	0.9038
8	0.3734	0.2978	0.8875	**0.9070**
9	0.3522	0.2944	0.8936	0.9066
10	0.3409	0.2958	0.8978	0.9030

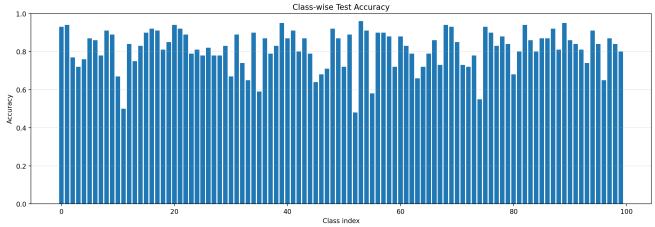
1.6 Q1 Discussion

Key Observations

1. Head-only baseline already achieves **81.43%** test accuracy — a strong starting point for CIFAR-100.
2. Within the fixed grid, rank ≥ 4 with larger α gives the best validation accuracy, suggesting moderate LoRA capacity is beneficial.
3. Optuna selects dropout $\approx$ 0.296 vs. fixed 0.1,

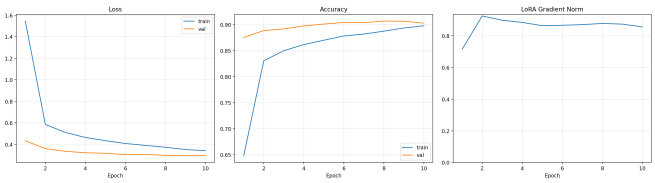


(a) Training curves

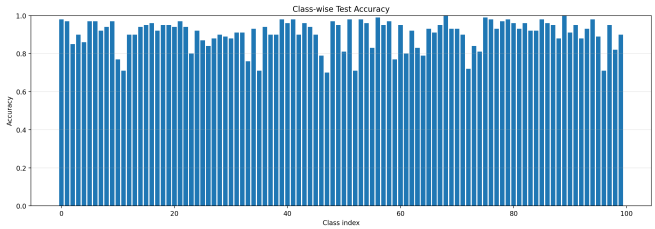


(b) Class-wise accuracy

Figure 1: Q1 baseline visualisations.



(a) LoRA training curves



(b) LoRA class-wise accuracy

Figure 2: Q1 best LoRA visualisations.

pushing val accuracy to **0.9070** through better regularisation.

4. All LoRA runs keep trainable parameters **under 1%** of the full model.

2. Q2: Adversarial Attacks & Detection

2.1 Objective

Question 2 has two parts: (i) compare a clean ResNet-18 against FGSM attacks implemented from scratch and via IBM ART; (ii) train ResNet-34 adversarial detectors for binary clean-vs-adversarial classification using PGD- and BIM-generated samples.

2.2 Q2 Clean Classifier

Table 5: Q2 clean ResNet-18 summary.

Metric	Value
Best Val Acc.	0.7878
Test Acc.	0.7836
Test Loss	0.6305
Meets 72% Target	✔ Yes

Table 6: Q2 clean classifier epoch metrics.

Ep.	Tr. Loss	Va. Loss	Tr. Acc	Va. Acc
1	1.6344	1.4078	0.4050	0.4902
2	1.3059	1.1641	0.5289	0.5912
3	1.1359	1.0362	0.5904	0.6338
4	1.0333	0.9186	0.6317	0.6648
5	0.9526	0.8893	0.6622	0.6748
6	0.8964	0.8562	0.6808	0.7020
7	0.8415	0.7931	0.7012	0.7158
8	0.8011	0.7758	0.7181	0.7230
9	0.7620	0.7795	0.7292	0.7258
10	0.7170	0.7177	0.7467	0.7384
11	0.6842	0.6796	0.7559	0.7600
12	0.6506	0.6903	0.7691	0.7536
13	0.6203	0.6704	0.7813	0.7620
14	0.5872	0.6390	0.7916	0.7716
15	0.5645	0.6283	0.8004	0.7786
16	0.5404	0.6162	0.8091	0.7828
17	0.5222	0.6106	0.8148	0.7808
18	0.5052	0.6048	0.8223	0.7874
19	0.4975	0.6043	0.8230	**0.7878**
20	0.4896	0.6010	0.8258	0.7858

2.3 FGSM: Scratch vs IBM ART

FGSM Perturbation Rule

$$x_{\text{adv}} = x + \epsilon \, \text{sign}\big(\nabla_x \mathcal{L}(\theta, x, y)\big)$$

Table 7: FGSM accuracy across perturbation strengths.

ϵ	Clean	Scratch	ART	Drop (Scratch)
0.01	0.7838	0.2341	0.3504	0.5497
0.03	0.7838	0.0196	0.1312	0.7642
0.05	0.7838	0.0065	0.1113	0.7773
0.10	0.7838	0.0090	0.0914	0.7748

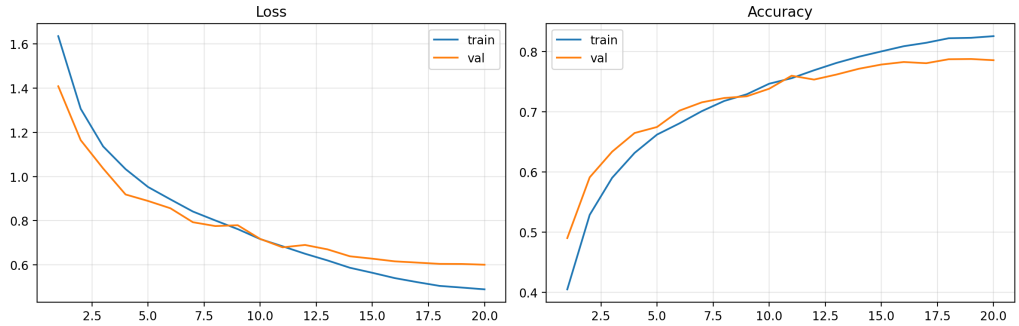


Figure 3: Clean ResNet-18 classifier training curves.

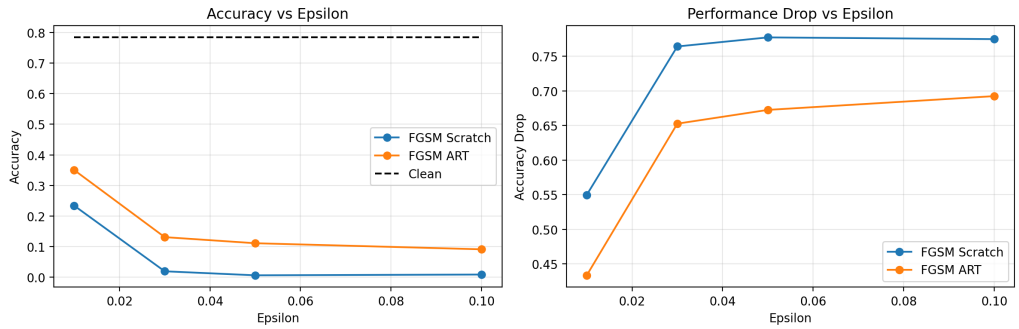


Figure 4: Perturbation strength ϵ vs. classifier accuracy.

Key Observations

1. Clean accuracy is **78.38%**; both FGSM variants collapse performance sharply even at ϵ=0.01.
2. The scratch implementation is slightly stronger than ART across all tested ϵ values in this setup.
3. As ϵ grows, adversarial accuracy approaches zero, confirming **first-order fragility** of the clean model.

Note

ART FGSM retains slightly higher accuracy at all ϵ, likely due to internal input clipping and normalisation differences.

2.4 Adversarial Detectors (ResNet-34)

Table 8: Detector summary.

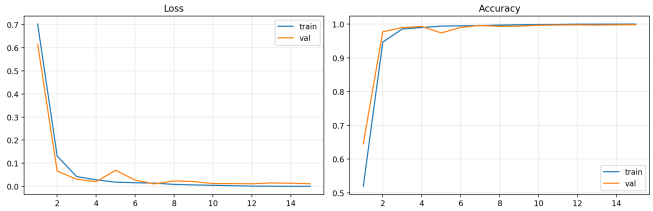
Attack	Val Acc.	Test Acc.	Loss	≥70%
PGD	0.9981	**0.9985**	0.0088	✓
BIM	0.9979	0.9983	0.0087	✓

Table 9: PGD detector epoch metrics.

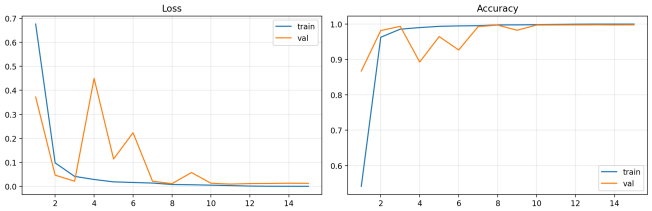
Ep.	Tr. Loss	Va. Loss	Tr. Acc	Va. Acc
1	0.7022	0.6155	0.5195	0.6456
2	0.1314	0.0657	0.9466	0.9772
3	0.0422	0.0308	0.9853	0.9897
4	0.0284	0.0201	0.9899	0.9930
5	0.0180	0.0695	0.9941	0.9738
6	0.0156	0.0269	0.9949	0.9901
7	0.0139	0.0105	0.9954	0.9960
8	0.0083	0.0235	0.9970	0.9934
9	0.0062	0.0209	0.9980	0.9938
10	0.0045	0.0119	0.9985	0.9965
11	0.0028	0.0119	0.9990	0.9972
12	0.0007	0.0109	0.9997	0.9977
13	0.0005	0.0148	0.9999	0.9971
14	0.0001	0.0136	0.99997	0.9978
15	0.0001	0.0110	0.99997	**0.9981**

Table 10: BIM detector epoch metrics.

Ep.	Tr. Loss	Va. Loss	Tr. Acc	Va. Acc
1	0.6755	0.3715	0.5413	0.8672
2	0.0975	0.0466	0.9628	0.9818
3	0.0414	0.0214	0.9856	0.9936
4	0.0285	0.4495	0.9901	0.8928
5	0.0186	0.1138	0.9936	0.9647
6	0.0160	0.2232	0.9948	0.9267
7	0.0136	0.0212	0.9955	0.9929
8	0.0077	0.0112	0.9976	0.9971
9	0.0066	0.0574	0.9977	0.9826
10	0.0049	0.0130	0.9983	0.9972
11	0.0031	0.0093	0.9990	0.9975
12	0.0011	0.0115	0.9996	0.9977
13	0.0003	0.0119	0.9999	0.9979
14	0.0002	0.0130	0.99994	0.9976
15	0.0001	0.0123	0.99997	**0.9978**



(a) PGD training



(b) BIM training

(c) PGD samples

(d) BIM samples

Figure 5: Detector training curves and qualitative sample comparisons.

2.5 Q2 Discussion

Key Observations

1. Clean ResNet-18 meets the 72% target at **78.36%** test accuracy.
2. FGSM sharply degrades clean-model accuracy, confirming **first-order adversarial fragility** of standard training.
3. Both ResNet-34 detectors achieve >**99.8%** test accuracy, far exceeding the 70% detection threshold.
4. BIM val loss spikes at epochs 4–6 (up to 0.4495), showing transient instability, yet converges to near-perfect by epoch 13.

3. Conclusion

In Question 1, parameter-efficient LoRA fine-tuning yields a **+9.06 pp** test accuracy gain over head-only fine-tuning on CIFAR-100, with Optuna further improving results by selecting a larger dropout — all while keeping trainable parameters below 1% of the full model. In Question 2, a standard ResNet-18 on CIFAR-10 collapses under even mild FGSM perturbations, yet dedicated ResNet-34 adversarial detectors for PGD and BIM achieve near-perfect binary classification accuracy, comfortably surpassing all assignment targets.